

Bahria University, Islamabad Campus

Department of Computer Science

Software Engineering Documentation Project

Assignment # 3

University-Based Smart Parking System

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Prepared for SEN 220 Assignment #3

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List of Acronyms

Acronym	Full Form
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
FR	Functional Requirement
GPS	Global Positioning System
NFR	Non-Functional Requirement
QR	Quick Response Code
RFID	Radio Frequency Identification
SEN	Software Engineering
SRS	Software Requirements Specification
UML	Unified Modeling Language
VCS	Version Control System

Abstract

This report presents the Software Engineering documentation for a University-Based Smart Parking System. The system is designed for a university campus where students, faculty, staff, visitors, security guards, and parking administrators need a controlled and organized method for managing parking spaces. The system supports vehicle registration, parking permit management, parking slot availability, slot reservation, gate entry and exit verification, occupancy monitoring, violation recording, reporting, and notifications.

The project is documentation-based. The objective is not to develop or deploy a working application, but to analyze and formally document the system using Software Engineering artefacts. The report includes problem statement, project scope, functional and non-functional requirements, use cases, use case description tables, context diagram, data flow diagrams, class diagram, sequence diagram, activity diagram, state machine diagram, layered architecture, database schema mapping, testing plan, sample test cases, version control usage, project planning, risk analysis, and team roles. The formatting style follows the provided sample report structure and the deliverables required in the SEN 220 assignment brief.

1. Introduction

Parking is a common operational problem in universities because a large number of students, faculty members, staff members, visitors, and service vehicles may enter the campus during the same time period. In a manual parking process, drivers may not know which parking zones have available space, security guards may take time to verify permits, and administrators may lack reliable data about occupancy, peak hours, violations, and parking demand.

The University-Based Smart Parking System is proposed as a digital parking management solution for a campus environment. It is intended to help drivers register vehicles, apply for permits, reserve spaces, check parking availability, and enter or leave campus parking areas through QR, RFID, or plate-based verification. Security guards use the system to validate vehicles, handle exceptions, and report violations. Administrators use the system to manage users, vehicles, permits, parking zones, slots, reports, and policies.

This report documents the Smart Parking System from a Software Engineering point of view. The focus is on reverse engineering, analysis, modeling, architecture, and testing documentation rather than programming. The system is described as if it were to be implemented in a real university, but the assignment deliverable remains a documentation project.

2. Project Objectives

- Document a real-world university parking management problem using Software Engineering concepts.
- Identify stakeholders, actors, requirements, constraints, and business rules for the Smart Parking System.
- Prepare complete requirement models including functional requirements, non-functional requirements, use cases, and use case description tables.
- Prepare structural and behavioral models including context diagram, DFD, class diagram, sequence diagram, activity diagram, and state machine diagram.
- Define a clear layered architecture that separates presentation, controller, business logic, data access, database, and external device responsibilities.
- Prepare a test plan, sample test cases, version control usage plan, Gantt chart, PERT chart, risk analysis, and two-member team role distribution.
- Present the documentation in a professional report format suitable for final submission.

3. Documentation Methodology

The documentation methodology follows a Software Engineering analysis process. First, the university parking problem is studied from the perspective of students, faculty members, visitors, guards, and administrators. Second, the system boundary is defined so that the core software system can be separated from external devices such as sensors, cameras, QR scanners, RFID readers, and gate barriers. Third, the main functions and quality requirements are documented. Fourth, diagrams are prepared to represent the system visually. Finally, testing and management documentation is prepared to show how the proposed system would be verified and controlled.

Because the assignment is based on documentation, no physical hardware, payment gateway, live database, or deployed application is implemented. Hardware components are treated as external actors or external interfaces. The report uses standard Software Engineering artefacts, including requirement tables, use case tables, DFDs, UML diagrams, database schema mapping, and test case tables.

3.1 Project Scope

The scope of the Smart Parking System includes managing campus parking for regular university users and visitors. The system covers vehicle registration, permit request and approval, slot availability, reservation, digital pass generation, gate entry and exit logging, occupancy monitoring, violations, notifications, and management reports.

In Scope	Out of Scope
User login and role-based access for students, faculty, staff, visitors, guards, and administrators.	Actual development of a complete mobile or web application.
Vehicle registration with plate number, owner, type, and color.	Physical installation of sensors, cameras, RFID readers, or gate barriers.
Permit request, approval, rejection, renewal, suspension, and expiry tracking.	Implementation of real-time ANPR or computer vision algorithms.
Parking zone/slot availability and reservation management.	Integration with government vehicle databases.
Gate check-in/check-out verification and entry/exit logs.	Live commercial payment gateway or fine payment system.

Security guard violation reporting and manual override logging.	Advanced AI-based parking prediction or route optimization.
Admin reports, planning, risk analysis, test cases, and VCS evidence plan.	Production deployment, server hosting, or hardware maintenance.

3.2 Stakeholders and Actors

Actor / Stakeholder	Description
Student / Faculty / Staff	Registered university users who register vehicles, apply for permits, reserve parking, check in/out, and view history.
Visitor	Temporary campus visitor who requests a visitor pass and uses authorized visitor parking areas.
Security Guard	Gate or parking-area staff member who verifies permits, handles exceptions, records manual overrides, and reports violations.
Parking Administrator	Responsible person who manages users, permits, zones, slots, violations, reports, and policies.
Sensor / Gate Device	External hardware such as QR scanner, RFID reader, ANPR camera, occupancy sensor, or gate barrier.
Notification Service	External email, SMS, or app notification service used for permit, reservation, expiry, and violation alerts.
University Management	Decision makers who use reports for parking planning, capacity decisions, and policy improvement.

4. Milestone 1: Requirements

This milestone documents the requirements of the Smart Parking System. It includes functional requirements, non-functional requirements, business rules, assumptions, constraints, use case diagram, and use case description tables.

4.1 Assumptions and Constraints

Type	Statement
Assumption	The university has defined parking zones such as student parking, faculty/staff parking, visitor parking, reserved parking, and disabled-accessible parking.
Assumption	Each student, faculty member, or staff member has a unique university identity. Visitors can be registered temporarily.
Assumption	Security guards have access to a guard interface or verification device at the gate.
Assumption	Parking sensors, QR scanners, RFID readers, ANPR cameras, and gate barriers are treated as external devices.
Constraint	This assignment is documentation-only; no working application or real hardware implementation is required.
Constraint	Real GitHub/Bitbucket screenshots must be added by the team after creating the repository, if required by the instructor.
Constraint	The system must remain understandable and suitable for a university-level Software Engineering assignment.

4.2 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow students, faculty, staff, visitors, security guards, and administrators to log in according to assigned roles.
FR-02	The system shall allow university users to register vehicles with plate number, vehicle type, color, owner name, and contact details.
FR-03	The system shall allow visitors to request temporary visitor parking passes.
FR-04	The system shall allow users to apply for a parking permit based on user role and university policy.

FR-05	The system shall allow the parking administrator to approve, reject, suspend, renew, or expire parking permits.
FR-06	The system shall display real-time or near-real-time availability of parking zones and slots.
FR-07	The system shall allow eligible users to reserve an available parking slot or parking zone for a limited time period.
FR-08	The system shall generate a digital parking pass or QR code for approved reservations and permits.
FR-09	The system shall verify a vehicle at the gate using permit status, reservation status, plate number, QR code, or RFID.
FR-10	The system shall record entry time, exit time, gate number, vehicle details, and verification result.
FR-11	The system shall update parking slot status as available, reserved, occupied, blocked, under maintenance, or closed.
FR-12	The system shall receive occupancy data from parking sensors or manual updates from guards/admins.
FR-13	The system shall allow security guards to perform manual gate override with a required reason.
FR-14	The system shall allow security guards to report parking violations such as wrong slot parking, expired permit, unauthorized parking, or overstay.
FR-15	The system shall notify users about permit approval/rejection, reservation confirmation, reservation expiry, violation, and gate denial.
FR-16	The system shall allow administrators to manage users, vehicles, permits, zones, slots, devices, violations, and reports.
FR-17	The system shall generate reports about occupancy, peak hours, permit usage, entry/exit logs, violations, and slot utilization.
FR-18	The system shall allow users to view parking history, active permit, active reservation, and violation status.

4.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	The system should be simple enough for students, visitors, guards,

		and administrators to use with minimum training.
NFR-02	Performance	Slot availability and gate verification should normally respond within 3 seconds under normal load.
NFR-03	Security	Passwords must be stored using secure hashing, and role-based access must protect admin and guard functions.
NFR-04	Availability	The system should remain available during university operating hours, especially during morning and evening peak times.
NFR-05	Reliability	Entry/exit logs, permit records, and violation records should not be lost during normal operation.
NFR-06	Scalability	The design should support additional users, parking zones, slots, gates, and sensors in the future.
NFR-07	Maintainability	The system should separate presentation, controller, business logic, data access, and database responsibilities.
NFR-08	Compatibility	The system should work on common desktop and mobile browsers used by university users.
NFR-09	Privacy	Vehicle, identity, and parking activity data should be visible only to authorized users.
NFR-10	Auditability	Sensitive actions such as permit approval, gate override, and violation closure should be logged.
NFR-11	Recoverability	The system should support regular backups of important records.
NFR-12	Accessibility	Screens should use readable text, clear labels, and simple navigation for different user groups.

4.4 Business Rules

ID	Business Rule
BR-01	Only approved university users or approved visitors can receive an active parking pass.
BR-02	One active permit can be linked with one primary registered vehicle unless the administrator allows multiple vehicles.
BR-03	A reserved slot must automatically expire if the vehicle does not check in within the allowed grace period.
BR-04	Faculty/staff parking zones cannot be reserved by students unless the administrator

	grants temporary access.
BR-05	A blocked, closed, or maintenance slot cannot be reserved or assigned.
BR-06	A manual gate override must include guard ID, reason, date/time, gate number, and vehicle plate number.
BR-07	Repeated violations may result in permit suspension according to university policy.
BR-08	Visitor passes must have a limited validity period and should expire automatically.

4.5 Use Case Diagram

The use case diagram shows the main interactions between users and the Smart Parking System. It includes regular campus users, visitors, guards, administrators, external devices, and notification services.

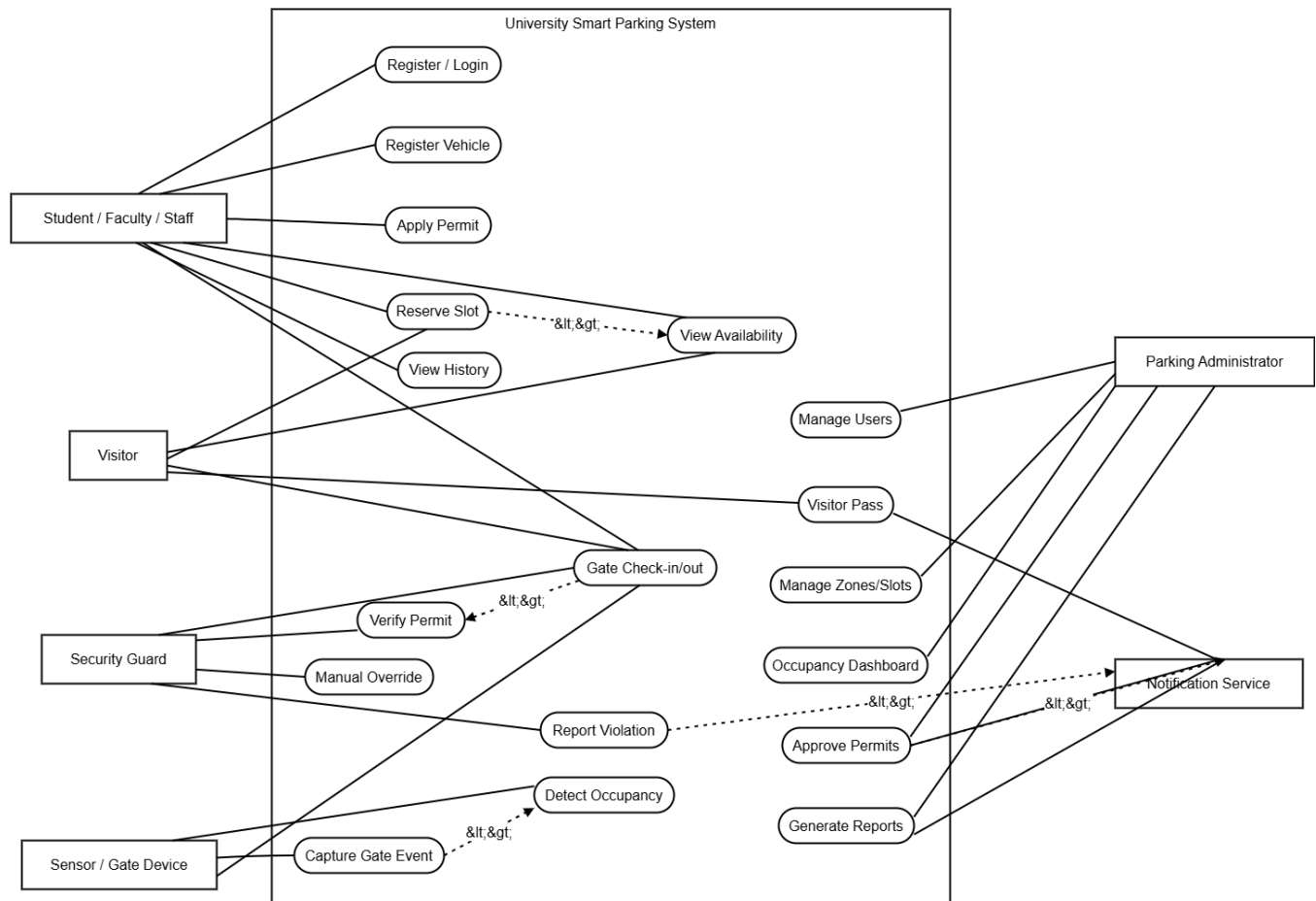


Figure 1 Use Case Diagram for University-Based Smart Parking System

4.6 Use Case Description Tables

UC-01 - Register / Login

Field	Description
Use Case ID	UC-01
Use Case Name	Register / Login
Primary Actor	Student/Faculty/Staff/Visitor/Admin/Guard
Goal	Access the system securely
Preconditions	User has valid identity or account details.
Main Flow	User opens the system, enters credentials, system verifies credentials, and role-based dashboard is displayed.
Alternative Flow	Invalid credentials show error; inactive accounts are blocked.
Postcondition	User session is created.

UC-02 - Register Vehicle

Field	Description
Use Case ID	UC-02
Use Case Name	Register Vehicle
Primary Actor	Student/Faculty/Staff
Goal	Add a vehicle for parking use
Preconditions	User is logged in.
Main Flow	User enters plate number, vehicle type, color and owner details; system validates and stores vehicle record.
Alternative Flow	Duplicate plate number or incomplete record shows validation error.
Postcondition	Vehicle record is stored.

UC-03 - Apply for Parking Permit

Field	Description
Use Case ID	UC-03
Use Case Name	Apply for Parking Permit
Primary Actor	Student/Faculty/Staff/Visitor
Goal	Request permission to park on campus

Preconditions	Vehicle is registered.
Main Flow	User selects permit type, attaches required details, submits request; system forwards request for approval.
Alternative Flow	Incomplete details return validation error; ineligible user is rejected.
Postcondition	Permit request is pending, approved, or rejected.

UC-04 - Reserve Parking Slot

Field	Description
Use Case ID	UC-04
Use Case Name	Reserve Parking Slot
Primary Actor	Student/Faculty/Staff/Visitor
Goal	Reserve an available slot or zone before arrival
Preconditions	User has active permit/pass.
Main Flow	User views availability, selects zone/slot, system checks permit and rules, then creates reservation.
Alternative Flow	If no slot is available, system suggests another zone or time.
Postcondition	Reservation is created and QR/pass is generated.

UC-05 - Gate Entry Verification

Field	Description
Use Case ID	UC-05
Use Case Name	Gate Entry Verification
Primary Actor	User, Security Guard, Gate Device
Goal	Allow only authorized vehicles to enter
Preconditions	Vehicle reaches gate.
Main Flow	Gate scans QR/Rfid/plate; system validates permit/reservation; barrier opens if valid.
Alternative Flow	Invalid or expired pass denies entry and sends alert to guard.
Postcondition	Entry log is created.

UC-06 - Update Slot Occupancy

Field	Description
Use Case ID	UC-06

Use Case Name	Update Slot Occupancy
Primary Actor	Sensor Device / Guard
Goal	Keep slot status accurate
Preconditions	Sensor/device is active or guard is authorized.
Main Flow	Sensor detects occupied/free state and sends update; system updates slot status.
Alternative Flow	If sensor fails, guard/admin manually updates slot.
Postcondition	Slot status is updated.

UC-07 - Report Parking Violation

Field	Description
Use Case ID	UC-07
Use Case Name	Report Parking Violation
Primary Actor	Security Guard
Goal	Record wrong or unauthorized parking
Preconditions	Guard observes violation.
Main Flow	Guard searches vehicle, selects violation type, adds notes/photo reference, and submits report.
Alternative Flow	If vehicle not found, temporary violation record is created.
Postcondition	Violation is saved and notification may be sent.

UC-08 - Manage Parking Zones and Reports

Field	Description
Use Case ID	UC-08
Use Case Name	Manage Parking Zones and Reports
Primary Actor	Parking Administrator
Goal	Control resources and review performance
Preconditions	Admin is logged in.
Main Flow	Admin adds/updates zones and slots, views dashboard, exports occupancy and violation reports.
Alternative Flow	System prevents deletion of zones with active reservations.
Postcondition	Parking data and reports are updated.

5. Milestone 2: Modeling

This milestone documents the system using structural, data-flow, behavioral, and interaction models. These diagrams explain how the Smart Parking System communicates with external entities, how data moves between internal processes, how domain classes are related, and how key workflows behave over time.

5.1 Context Diagram

The context diagram treats the Smart Parking System as one complete process. It shows the system boundary and interactions with users, guards, administrators, devices, and notification services.

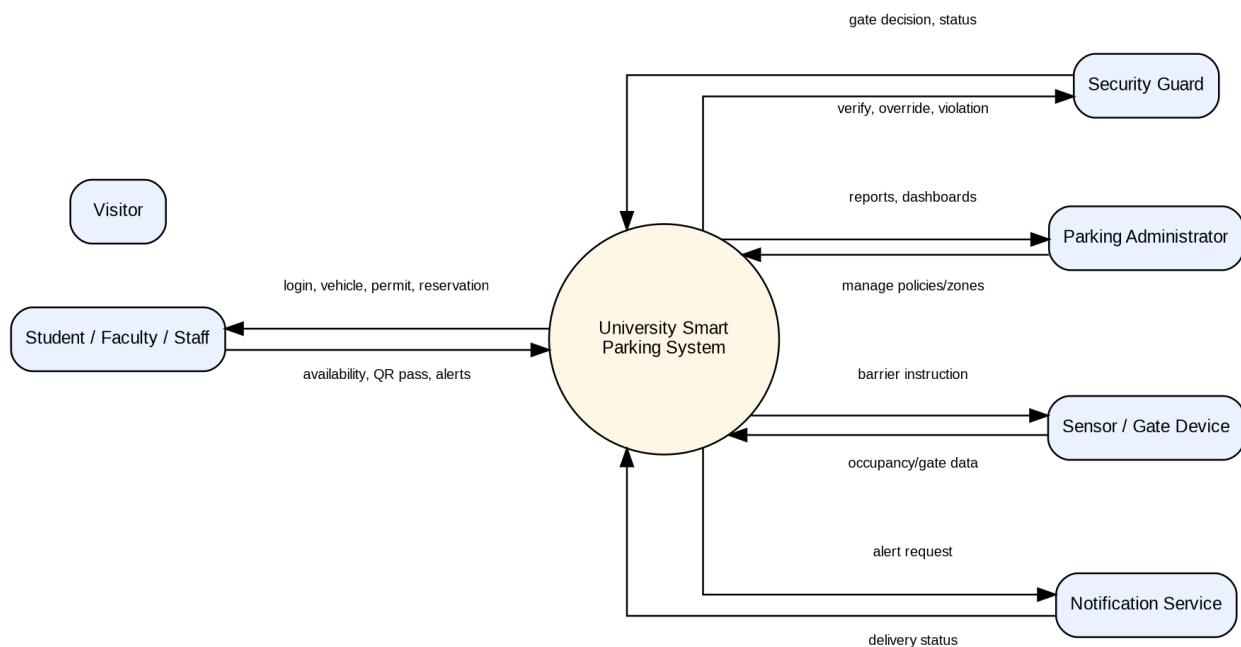


Figure 2 Context Diagram

5.2 DFD Level 0

The Level 0 DFD decomposes the system into major processes such as authentication, permits, zones, reservations, gate entry/exit, occupancy monitoring, violations, and reporting.

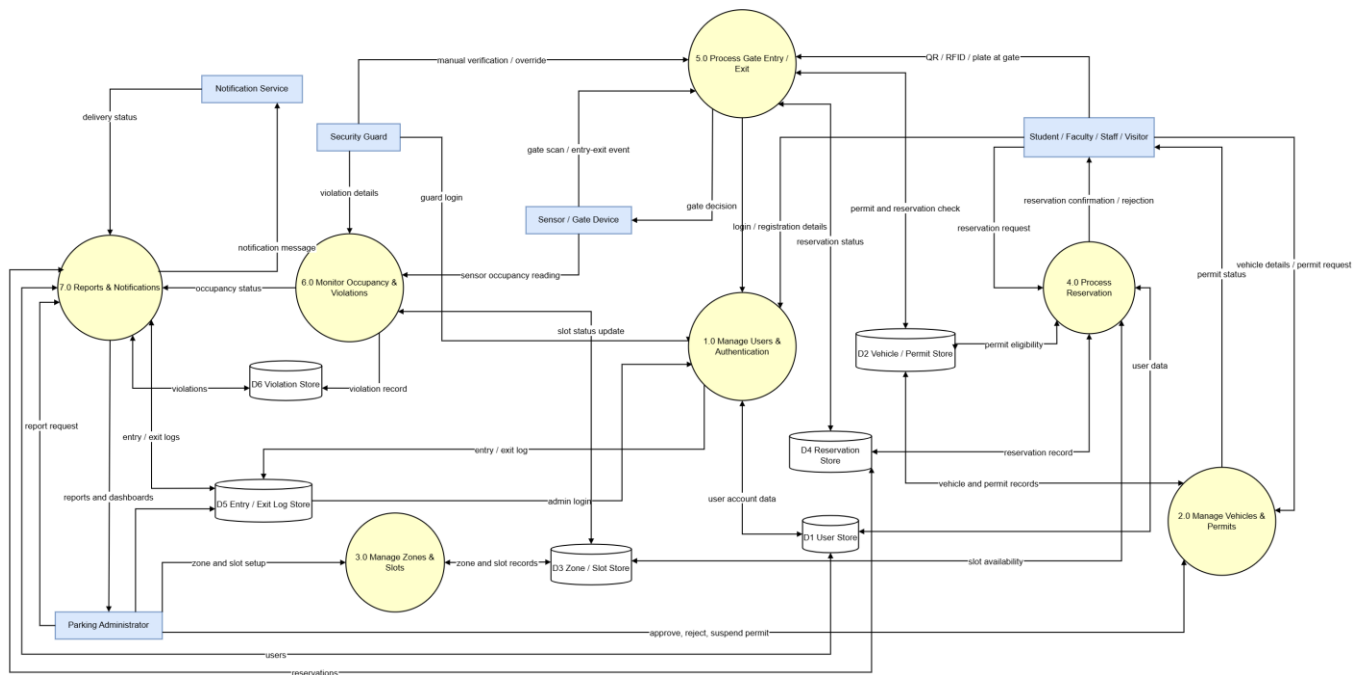


Figure 3 DFD Level 0

5.3 DFD Level 1 - Reservation Process

The Level 1 DFD expands the reservation process into receiving a request, validating the user and vehicle, checking permit eligibility, checking slot availability, creating the reservation, and sending confirmation.

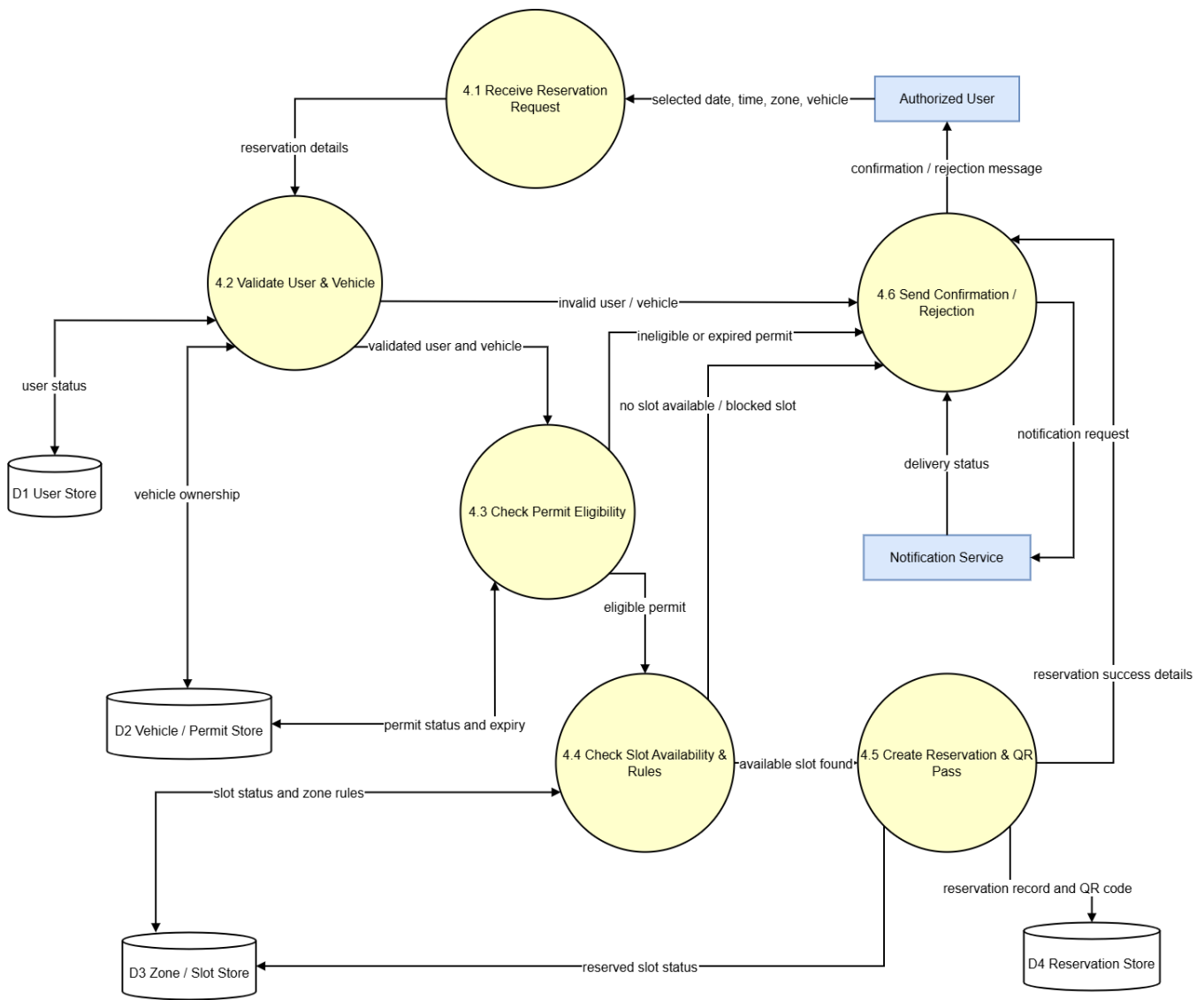


Figure 4 DFD Level 1 - Reservation Process

5.4 Class Diagram

The class diagram identifies the main domain entities and relationships. It is useful for understanding how the database and object model can be structured.

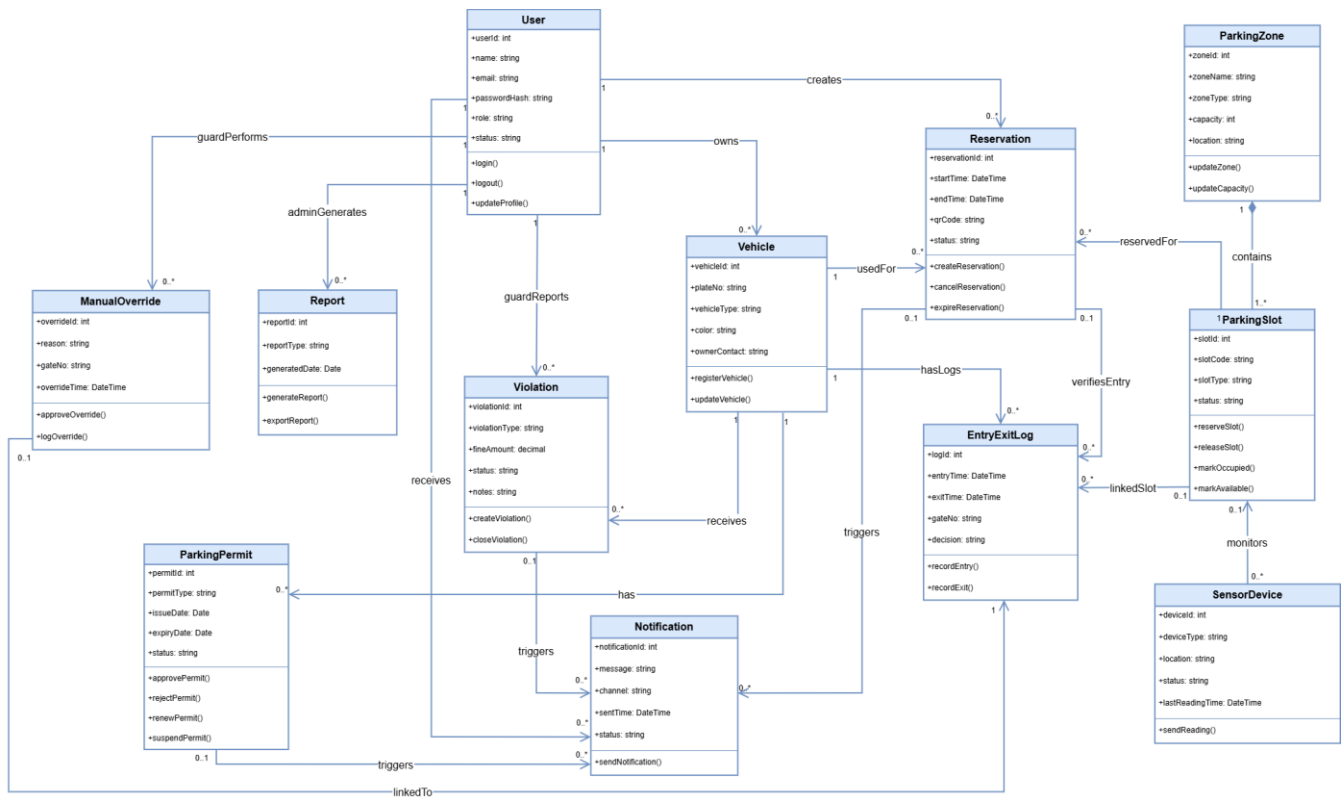


Figure 5 UML Class Diagram

5.5 Sequence Diagram - Reserve and Enter Parking

The sequence diagram shows how a user reserves a parking slot and then uses the generated pass at the gate. It highlights interactions between the app, permit service, slot service, gate device, and entry/exit log.

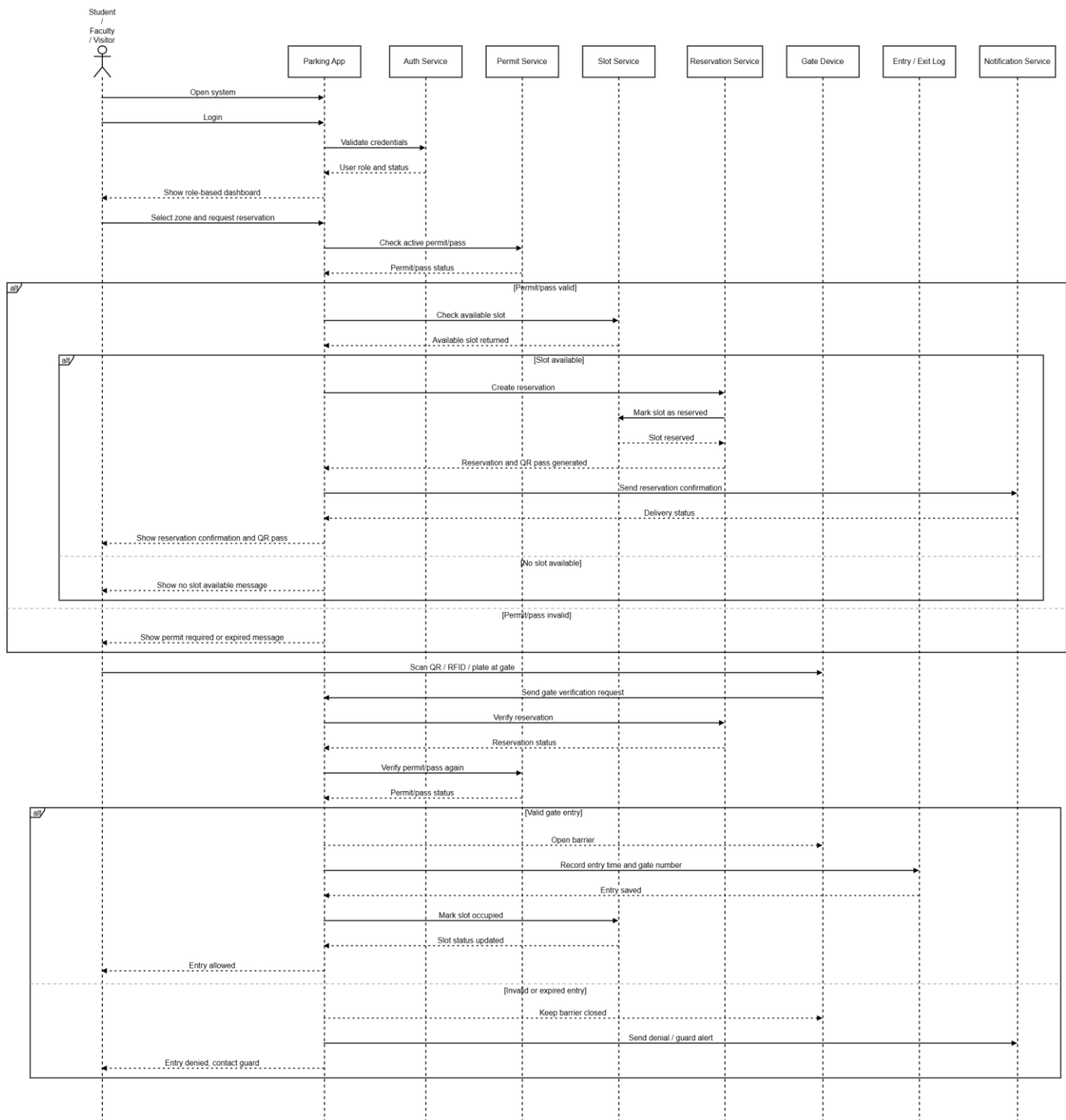


Figure 6 Sequence Diagram - Reserve and Enter Parking

5.6 Activity Diagram

The activity diagram describes the end-to-end workflow from login and vehicle registration to permit approval, reservation, gate verification, parking, and exit.

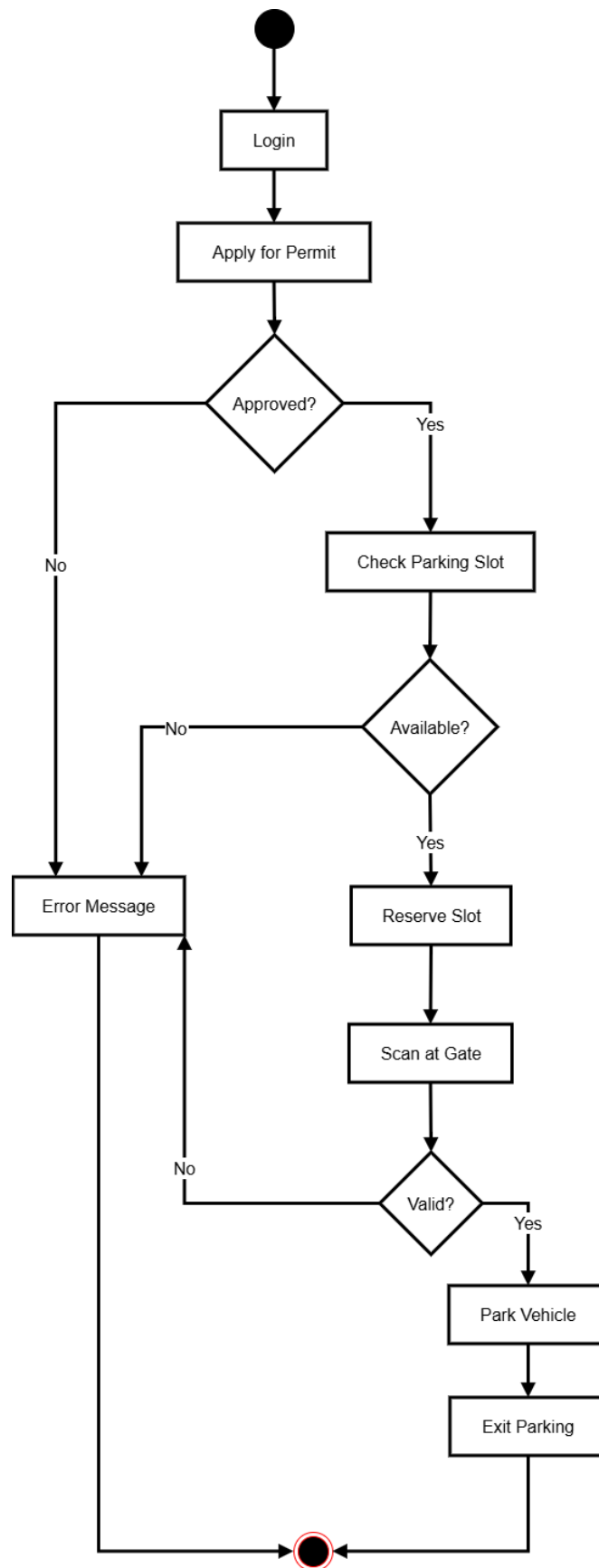


Figure 7 Activity Diagram

5.7 State Machine Diagram - Parking Slot

The state machine diagram shows how a parking slot changes between available, reserved, occupied, violation, maintenance, closed, and final states.

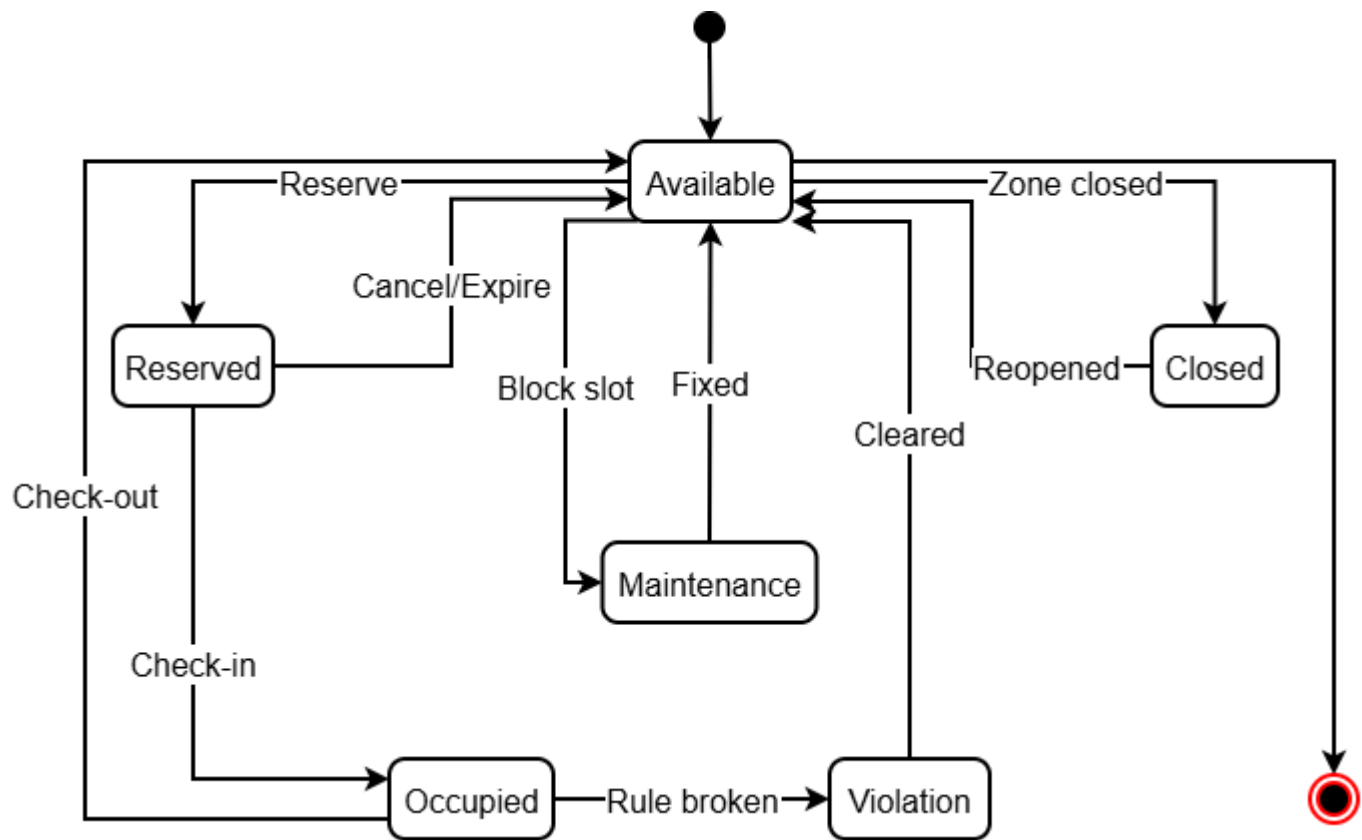


Figure 8 State Machine Diagram - Parking Slot

6. Milestone 3: Design and Architecture

This milestone refines the system design and defines the software architecture. The Smart Parking System is organized using a layered architecture. This keeps user interface, request handling, business rules, database access, and external device communication separate and easier to maintain.

6.1 Refined UML Design Explanation

The refined design separates user roles from domain entities. A user can own one or more vehicles. A vehicle can have one or more permit records over time. A user can create reservations, and a reservation is linked to a vehicle and parking slot. Slots belong to parking zones, and each slot may receive status updates from a sensor device. Entry and exit events are recorded separately so that administrators can analyze traffic and peak hours. Violations are linked to vehicles and may be created by guards. Reports are generated by administrators from stored records.

This design supports maintainability because core entities such as User, Vehicle, ParkingPermit, ParkingSlot, Reservation, EntryExitLog, and Violation have clear responsibilities. It also supports future expansion because new devices, new parking zones, or new reporting types can be added without changing the entire system.

6.2 Layered Architecture

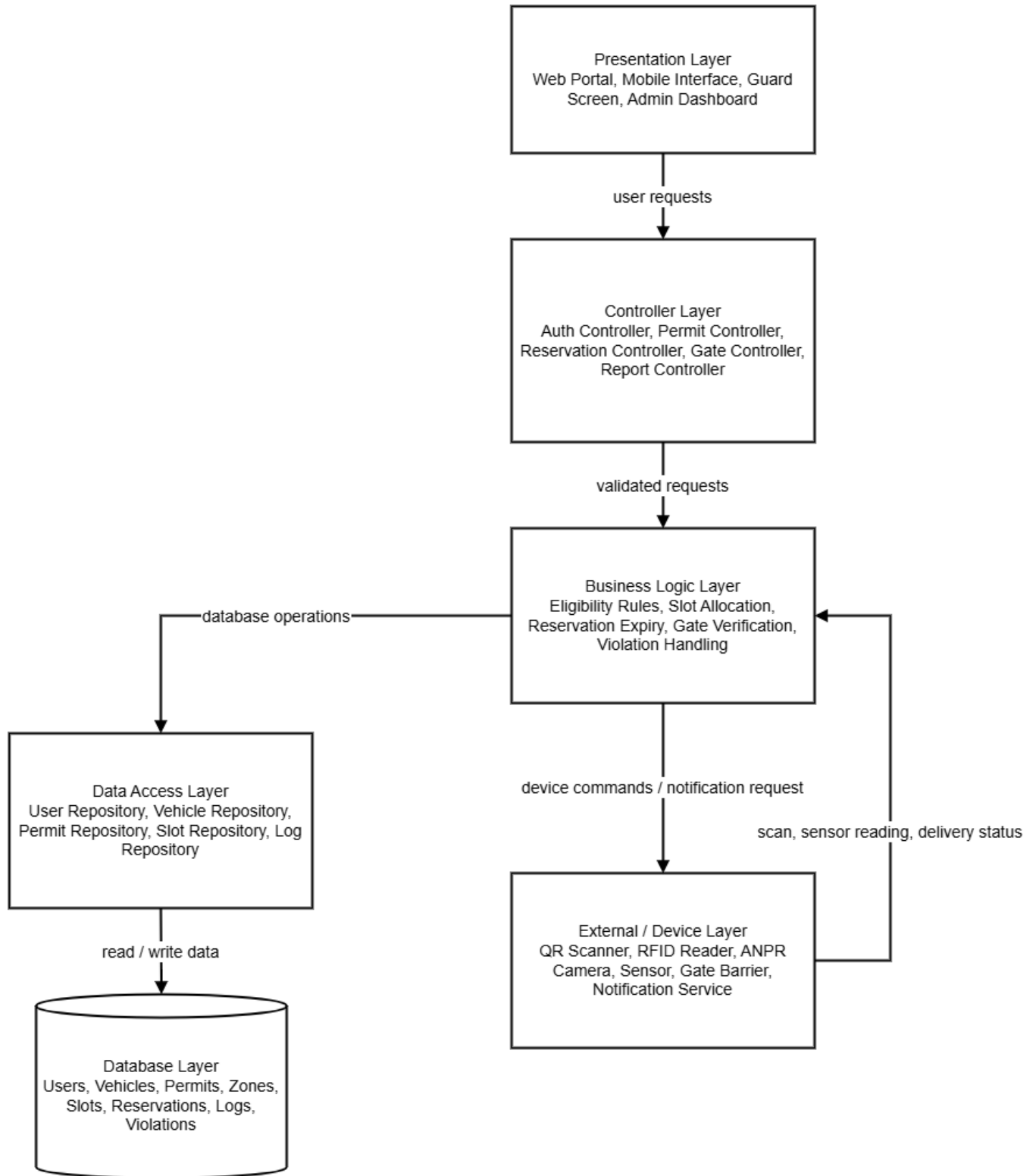


Figure 9 Layered Architecture of the Smart Parking System

Layer	Responsibility
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Presentation Layer	Provides the web portal, mobile interface, guard verification screen, and administrator dashboard. It is the part visible to end users.
Controller Layer	Receives user actions and routes them to the correct services such as authentication, permit, reservation, gate, violation, and report controllers.
Business Logic Layer	Contains core rules for permit eligibility, slot allocation, reservation expiry, gate verification, violation handling, and notification decisions.
Data Access Layer	Provides repository classes and database queries for storing and retrieving users, vehicles, permits, slots, reservations, logs, and violations.
Database Layer	Stores persistent data including users, vehicles, permits, parking zones, parking slots, reservations, entry/exit logs, violations, and generated reports.
External / Device Layer	Represents QR scanners, RFID readers, ANPR cameras, occupancy sensors, gate barriers, and notification services.

6.3 Database Schema Mapping

Table	Fields and Keys
User	user_id: int PK; name: varchar; email: varchar UNIQUE; password_hash: varchar; role: varchar; status: varchar
Vehicle	vehicle_id: int PK; user_id: int FK; plate_no: varchar UNIQUE; vehicle_type: varchar; color: varchar; owner_contact: varchar
ParkingPermit	permit_id: int PK; vehicle_id: int FK; permit_type: varchar; issue_date: date; expiry_date: date; status: varchar
ParkingZone	zone_id: int PK; zone_name: varchar; zone_type: varchar; capacity: int; location: varchar
ParkingSlot	slot_id: int PK; zone_id: int FK; slot_code: varchar; slot_type: varchar; status: varchar
Reservation	reservation_id: int PK; user_id: int FK; vehicle_id: int FK; slot_id: int FK; start_time: datetime; end_time: datetime; qr_code: varchar; status: varchar
EntryExitLog	log_id: int PK; vehicle_id: int FK; reservation_id: int FK NULL; entry_time: datetime; exit_time: datetime; gate_no: varchar; decision: varchar
Violation	violation_id: int PK; vehicle_id: int FK; guard_id: int FK; violation_type: varchar; fine_amount: decimal; status: varchar; notes: text
SensorDevice	device_id: int PK; slot_id: int FK NULL; device_type: varchar; location:

	varchar; status: varchar; last_reading_time: datetime
Notification	notification_id: int PK; user_id: int FK; message: text; channel: varchar; sent_time: datetime; status: varchar

6.4 Security and Validation Considerations

- Role-based access control should prevent students or visitors from accessing administrator and guard functions.
- Passwords should be stored as hashes instead of plain text.
- Vehicle plate numbers and user email addresses should be unique.
- Reservation creation should check permit validity, slot status, time conflicts, and user eligibility.
- Manual gate overrides should always be logged with guard ID, reason, vehicle plate number, gate number, and date/time.
- Reports containing personal or vehicle information should be visible only to authorized administrators.
- Input validation should be applied to vehicle plate numbers, email addresses, date/time fields, and permit request details.

7. Milestone 4: Testing and Management

This milestone documents how the Smart Parking System would be tested and managed if it were implemented. Because the assignment is documentation-based, the tests are planned sample tests rather than executed test results. The section also includes version control usage, project planning, risk analysis, and two-member team roles.

7.1 Test Plan

Testing Type	Purpose	Examples
Unit Testing	Check individual functions or modules.	Login validation, vehicle registration validation, permit validation, slot status update, QR pass generation.
Integration Testing	Check whether connected modules work together.	Reservation with permit validation, gate verification with reservation, sensor update with slot status.
System Testing	Check the complete end-to-end workflow.	User registers vehicle, obtains permit, reserves slot, checks in at gate, parks, and exits.
Usability Testing	Check whether the system is easy to use for each role.	Student reserves slot; guard verifies vehicle; admin generates report.
Security Testing	Check authorization and data protection.	Student cannot approve permits; guard cannot access admin reports; invalid login is rejected.
Performance Testing	Check response time under expected campus load.	Slot availability and gate verification should respond within 3 seconds.

7.2 Sample Test Cases

Test ID	Scenario	Input	Expected Output	Type	Status
TC-01	Login with valid university credentials	Valid email and password	Role-based dashboard is displayed	Unit	Planned
TC-02	Login with invalid password	Valid email and wrong password	Error message is shown and access is denied	Unit	Planned
TC-03	Register vehicle	Plate number,	Vehicle record is saved	Unit	Planned

	with complete information	type, color, owner details			
TC-04	Register duplicate plate number	Existing plate number	System rejects duplicate vehicle	Unit	Planned
TC-05	Apply for parking permit	Registered vehicle and permit type	Permit request is created as pending	Integration	Planned
TC-06	Admin approves permit	Pending permit request	Permit status changes to approved	Integration	Planned
TC-07	View parking availability	User selects parking zone	Available, reserved, occupied slots are displayed	System	Planned
TC-08	Reserve available slot	Valid permit and available slot	Reservation is created and QR pass is generated	Integration	Planned
TC-09	Reserve blocked slot	Blocked or maintenance slot	Reservation is rejected	Unit	Planned
TC-10	Enter gate with valid QR code	Valid active reservation	Gate opens and entry log is saved	System	Planned
TC-11	Enter gate with expired QR code	Expired reservation	Gate remains closed and guard alert is generated	System	Planned
TC-12	Sensor marks slot occupied	Sensor reading = occupied	Slot status changes to occupied	Integration	Planned
TC-13	Guard reports violation	Vehicle, violation type, notes	Violation record is saved and notification is sent	Integration	Planned
TC-14	Admin generates occupancy report	Date range and zone selected	Report is displayed/exported	System	Planned
TC-15	Unauthorized role access	Student opens admin page	Access denied message is displayed	Security	Planned

7.3 Version Control Usage

The team will use GitHub or Bitbucket as the version control system for managing the documentation project. Since the assignment is documentation-based, the repository will contain the final report, diagram files, planning files, references, and evidence screenshots. The following repository structure and commit history can be used as evidence in the report.

Repository Item	Description
Repository Name	sen220-smart-parking-documentation
README.md	Short project overview, members, scope, and deliverables.
report/	Final DOCX and PDF report files.
diagrams/	Use case, DFD, class, sequence, activity, state, architecture, and PERT diagram images.
planning/	Gantt chart, risk analysis, team roles, and task plan.
references/	References used for the documentation.

Commit No.	Commit Message	Responsible Member
1	Initial repository created with folder structure	Member 1
2	Added problem statement, scope, objectives, and stakeholders	Member 1
3	Added functional and non-functional requirements	Member 1
4	Added use case diagram and use case description tables	Member 2
5	Added context diagram, DFD Level 0, and DFD Level 1	Member 2
6	Added UML class, sequence, activity, and state machine diagrams	Member 2
7	Added architecture, database schema mapping, and security considerations	Member 1
8	Added test plan, sample test cases, Gantt chart, PERT chart, and risk analysis	Member 2
9	Final proofreading, formatting, and references added	Both Members

Evidence to attach before final upload if required by the instructor: repository homepage screenshot, commit history screenshot, folder structure screenshot, and final report file opened in the repository.

7.4 Project Planning Report

The project was planned for the period from 18-04-2026 to 10-05-2026. Tasks are divided between four members while All members review the final report together.

No.	Task	Start	End	Duration	Responsible
1	Select case study and confirm scope	18-04-2026	19-04-2026	2 days	All Members
2	Study university parking workflows	20-04-2026	21-04-2026	2 days	Zia Ullah
3	Write introduction, problem statement and objectives	22-04-2026	23-04-2026	2 days	Zia Ullah
4	Write requirements and business rules	24-04-2026	25-04-2026	2 days	Zia Ullah
5	Prepare use case diagram and use case tables	26-04-2026	27-04-2026	2 days	Abdul Ahad
6	Prepare context diagram and DFDs	28-04-2026	29-04-2026	2 days	Zaryab Ishaq
7	Prepare UML class, sequence, activity and state diagrams	30-04-2026	02-05-2026	3 days	Abdul Ahad
8	Prepare architecture and database schema mapping	03-05-2026	04-05-2026	2 days	Zaryab Ishaq
9	Prepare testing plan and test cases	05-05-2026	06-05-2026	2 days	Rana Bilal
10	Prepare VCS evidence, risk analysis and team roles	07-05-2026	08-05-2026	2 days	Rana Bilal
11	Final proofreading and formatting	09-05-2026	10-05-2026	2 days	All Members

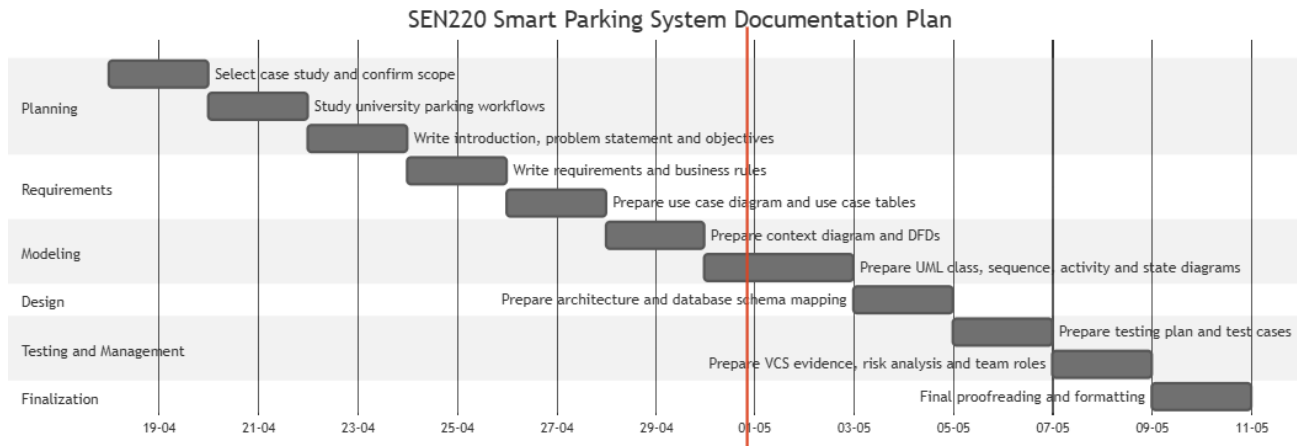


Figure 10 PERT Chart for Documentation Project

7.5 Risk Analysis

Risk ID	Risk	Probability	Impact	Mitigation
R-01	Assignment deliverables may be missed	Medium	High	Use checklist against milestone requirements before submission.
R-02	Diagrams may not match written requirements	Medium	High	Cross-check each diagram with functional requirements and use cases.
R-03	Only two members may face workload pressure	Medium	Medium	Divide work clearly and review each other's sections.
R-04	Plagiarism risk due to copied online material	Low	High	Write in original wording and add references.
R-05	GitHub evidence may be missing	Medium	Medium	Create repository early and take screenshots after each major update.
R-06	Formatting errors in final document	Medium	Medium	Use consistent headings, captions, tables, and final proofreading.
R-07	Incorrect DFD or	Medium	Medium	Keep diagrams simple, readable, and

	UML notation			aligned with SE concepts.
R-08	Ambiguous scope because hardware is involved	Medium	Medium	Treat hardware as external devices and document interfaces logically.

7.6 Team Roles - Four Members

Member	Role	Main Responsibilities
Muhammad Zia Ullah	Documentation & Requirements Lead	Prepare title page, abstract, introduction, objectives, problem statement, project scope, stakeholders, functional and non-functional requirements, business rules, and ensure final formatting consistency.
Zaryab Ishaq	System Design & Architecture Lead	Prepare architecture explanation, database schema mapping, security considerations, context diagram, and Data Flow Diagrams (DFDs).
Abdul Ahad Razaq	Modeling & Diagram Lead	Prepare use case diagram, use case description tables, class diagram, sequence diagram, activity diagram, state machine diagram, and diagram appendix.
Rana Muhammad Bilal	Testing & Project Planning Lead	Prepare test plan, sample test cases, Version Control System (VCS) usage (GitHub), Gantt/PERT chart, and risk analysis
All Members	Final Review Team	Review the full report, check alignment with assignment brief, verify diagrams, correct formatting, add GitHub evidence, and prepare final submission file.

References

1. Bahria University Islamabad Campus, Department of Computer Science. SEN 220 Software Engineering Assignment #3 Brief, spring 2026.
2. Sommerville, I. Software Engineering. Pearson Education.
3. Pressman, R. S. and Maxim, B. R. Software Engineering: A Practitioner's Approach. McGraw-Hill.
4. Fowler, M. UML Distilled: A Brief Guide to the Standard Object Modeling Language. Addison-Wesley.
5. IEEE Computer Society. IEEE Recommended Practice for Software Requirements Specifications.
6. Object Management Group. Unified Modeling Language Specification.
7. General observation of university parking workflows, gate verification, digital permits, QR-based access control, and parking management dashboards.

The End